

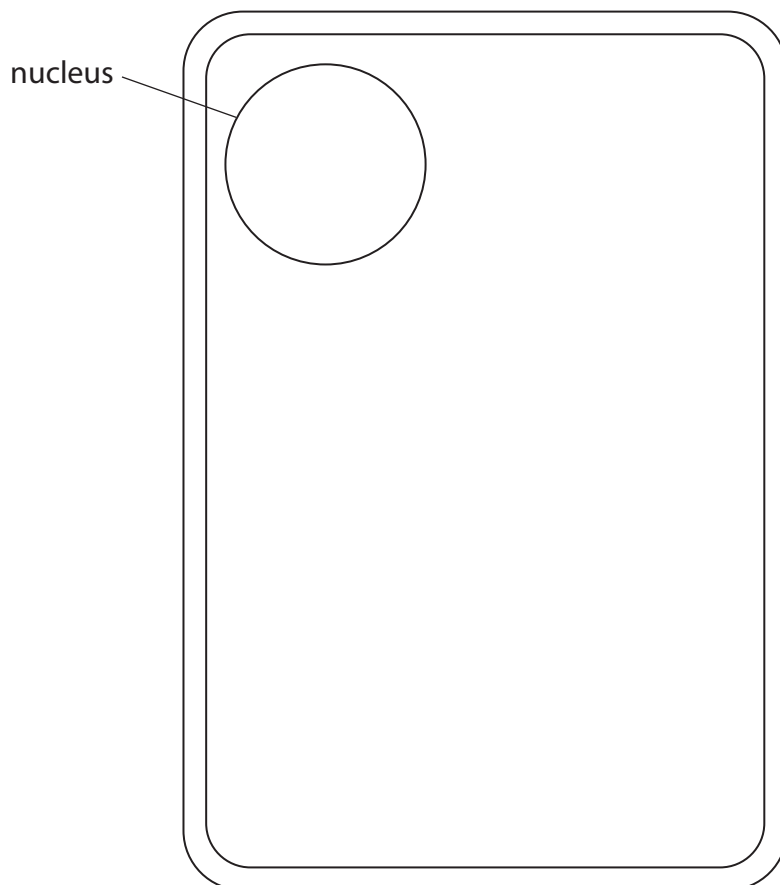
Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1** Woese classified organisms into a three-domain system.

Plant cells were classified into the Eukarya domain.

The diagram shows an incomplete cell from a plant.



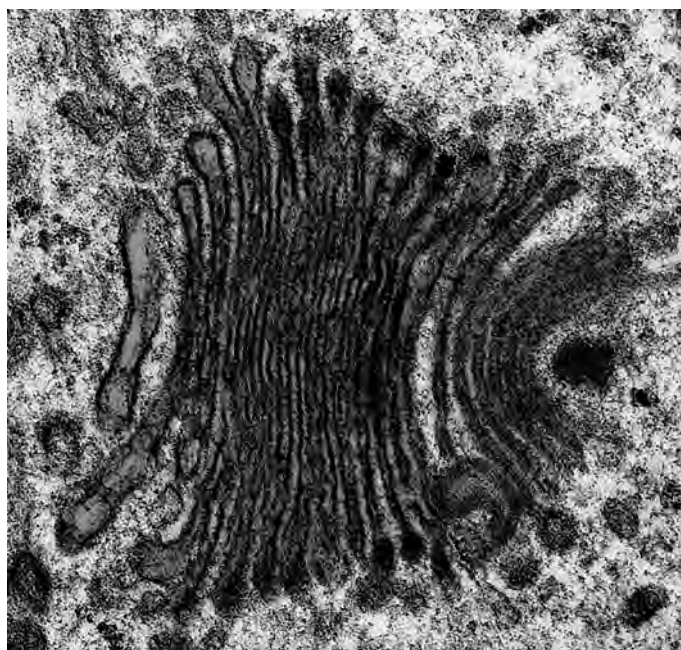
- (a) Complete the diagram by drawing an amyloplast, a chloroplast and the tonoplast to show their location and relative size in this cell.

Label these three structures.

(3)



- (b) The photograph shows an organelle found in a plant cell, as seen using an electron microscope.



(Source: © SCIENCE PHOTO LIBRARY)

State the function of this organelle.

(1)

- (c) The other domains in the three-domain system are Archaea and Bacteria. Complete the table to show in which of the domains the following structures would be found.

(4)

Structure	Archaea only	Bacteria only	Eukarya only	More than one domain
cell membrane	☐	☐	☐	☐
nucleolus	☐	☐	☐	☐
cell wall	☐	☐	☐	☐
70S (small) ribosome	☐	☐	☐	☐

(Total for Question 1 = 8 marks)